Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | *Subject: Computer Networks* | *Time: 03 Hours*

Code: 155602 | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option for the following questions (Answer any seven):*** **[2 x 7 = 14]**

(a) A network topology that needs a hub or central controller?

(i) Bus (iii) Ring

(ii) Star (iv) Mesh

(b) Which routing algorithm is used in RIP?

(i) Link state (iii) Distance vector

(ii) Path vector (iv) Hybrid

(c) CRC uses

(i) XOR (iii) AND

(ii) OR (iv) NOT

(d) CSMA/CD is used in

(i) Wi-Fi (iii) Ethernet

(ii) Bluetooth (iv) Fiber optic

(e) HTTP default port is

(i) 20 (iii) 21

(ii) 80 (iv) 110

(f) Which of the following protocols is the bit-oriented protocol?

(i) SSL (iii) HTTP

(ii) HDLC (iv) All of these

(g) ARP is used for mapping

(i) IP → MAC (iii) MAC → IP

(ii) Port → IP (iv) DNS → IP

(h) TCP is a

(i) Connectionless protocol (iii) Unreliable transport

(ii) Connection-oriented protocol (iv) Datagram protocol

(i) In congestion control, leaky bucket algorithm controls

(i) Flow rate (iii) Header size

(ii) Queue size (iv) Window size

(j) IPv4 address size is

(i) 16 bits (iii) 32 bits

(ii) 64 bits (iv) 128 bits

Q.2 (a) Compare the performance characteristics of coaxial, twisted pair and fiber optic transmission media. **[7]**

(b) What is LAN? Explain wired LAN, wireless LAN, and VLAN in detail. **[7]**

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Q.3	(a)	Draw the OSI model. State the responsibilities of each layer.	[7]
	(b)	Explain FDM, TDM, and WDM with suitable diagrams.	[7]
Q.4	(a)	Define bit rate and baud rate. An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?	[7]
	(b)	Explain how token ring technology is used to design LAN? In a token ring network the transmission speed is 10^7 bps and the propagation speed is 200 metres/microsecond. How much length of cable is equivalent to the 1-bit delay in this network?	[7]
Q.5	(a)	How Slotted ALOHA protocol differs from pure ALOHA? Write the vulnerable time and maximum throughput for both the protocols.	[7]
	(b)	Difference between the Go-Back-N ARQ and Selective Repeat ARQ sliding window protocol.	[7]
Q.6	(a)	Why DNS is needed? How does it work?	[7]
	(b)	Differentiate between TCP and UDP Protocol.	[7]
Q.7	(a)	What are the different techniques to improve Quality of Service. Explain Leaky Bucket algorithm for fixed size and variable length packets.	[7]
	(b)	How DHCP protocol provides static and dynamic address allocation?	[7]
Q.8	(a)	State the major difference between distance vector routing and link state routing. How these routing techniques work?	[7]
	(b)	Compare the Network layer protocols IPv4 and IPv6.	[7]

Q.9 Write Short notes on ***any two*** of the following: **[7 x 2 = 14]**

(a) Public & Private Key Cryptography
(b) Error Detection and Correction techniques
(c) CSMA/CD VS. CSMA/CA
(d) Network Topologies

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